[Question 3]

How much is a page of phuzics research worth as measured by the resulting discounted stream of academic salary?

1. The model

$$\log\left(\frac{S_{it}}{S_{i\,t-1}}\right) = \beta_0 + \beta_1 \log\left(\frac{Y_{it}}{Y_{i\,t-1}}\right) + \beta_2 RU_{it} + \beta_3 SEX_i + \alpha_i + u_{it},$$

S_{it} = salary of i-th person at time t,

Y_{it} = discounted cumulative research pages,

RU_{it} = dummy for employment in research university (= 1 if employed)

SEX_i = gender dummy, which is 0 (male) or 1 (female).

2. Estimation Results

-5 Estimation Strategy

: Pooled OLS, Fixed Effect Estimator, Between Estimators, Random Effect Estimators, Hausman-Taylor

-To test for whether the time-variant covariates are correlated with the individual effects, we used the Hausman specification test, and the value of test statistic,19.84, implies that we need to use the Hausman-Taylor approach in order to obtain consistent estimates

-Covariates are $X_1 = \log(\frac{Y_{it}}{Y_{i\,t-1}})$, $X_2 = RU_{it}$, and $Z_1 = SEX_i$.

X_2 is treated as endogenous. Therefore, [QX_1 QX_2 Z_1 PX_1] is used as a set of instrument variables for the Hausman-Taylor approach.

Table. Estimation of Salary Model

	Pooled OLS	Fixed	Between	Random	Hausman-Taylor
	coef/se	coef/se	coef/se	coef/se	coef/se
$\log(\frac{Y_{it}}{Y_{i\,t-1}})$	0.0524***	0.0507***	0.0547***	0.0517***	0.0524***
	(0.002)	(0.002)	(0.003)	(0.002)	(0.002)
RU_{it}	0.0084***	0.0044***	0.0086***	0.0078***	0.0045***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
SEX_i	-0.0055***	dropped	-0.0065***	-0.0057***	-0.0058***
	(0.001)		(0.001)	(0.001)	(0.002)
Cons	0.0145***	0.0161***	0.0144***	0.0152***	0.0167***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)

Note: *** p<0.01, ** p<0.05, * p<0.1

3. **The Worth of a Page of Phuzics Research**

-The representative phuzicists
(a) A male phuzicist from the PhD program ranked 2nd. (b) He is working in a research position with an initial paper of 4 pages and has the starting salary of $60,000 and his career spans 50 years.

-Research productivity model is used to estimate the corresponding cumulative discounted output profile

- The present value of the lifetime salary evaluated in the year of the PhD is as following

$$w_M = \sum_{t=1}^{T} \frac{S_t}{(1+r)^t}$$, where T=50 and r=0.03.

-Lifetime salary of the representative phuzicist is <u>$3 million.</u>

-This is equivalent to the annuity of <u>$113,254</u> per year at the notional 3% interest rate.

-By replacing y_p with $y_p + 1$, we can measure the adding one page of "innovation" in research output occurring *p* years after the PhD.

-Producing one more page of research three years after PhD is worth $420. However, the value of an additional page continues to decrease up to $14 at the end of his career.
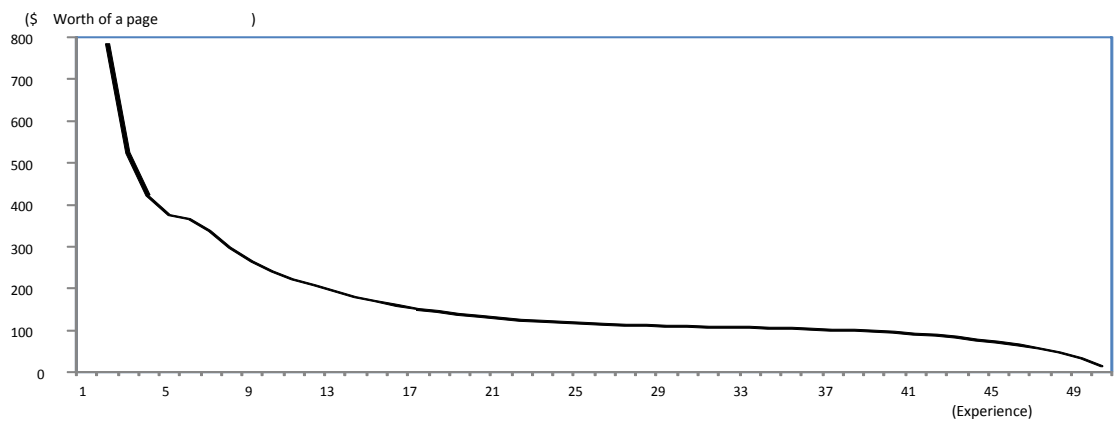
Figure. How much is a page of phuzics research worth?
$(RU_{it} = 1,\ SEX_i = 0, S_{it} = \$60{,}000)$